

HPS 220 - History and Philosophy of Science

Professor: Dr. Maya Patel | mpatel@university.edu | Humanities 318

Office Hours: Tue 12:00-1:30 PM, Thu 11:30 AM-1:00 PM, or by appointment

Term: Fall 2026 | Credits: 3

Location: See schedule below

Course Description

How does science work? What makes a theory scientific? How do scientific revolutions happen, and who gets to participate in them? This course surveys major themes in the history and philosophy of science from the Scientific Revolution to the present. We examine foundational texts by Kuhn, Popper, Lakatos, Feyerabend, and Longino, and apply their frameworks to case studies in physics, biology, and medicine. Discussion-heavy format: students are expected to come to every class having done the readings. The course meets Tuesday and Thursday from 10:00-11:15 AM in Humanities 115.

Learning Objectives

Students will be able to:

1. Explain and critically evaluate major philosophical accounts of scientific method.
2. Analyze historical episodes of scientific change using Kuhn's and Lakatos's frameworks.
3. Situate science within its social, political, and cultural context.
4. Write clear, well-argued philosophical essays using textual evidence.
5. Engage charitably with opposing viewpoints in seminar discussion.

Required Materials

All readings provided as PDFs on the course portal (no textbook purchase required).

Key texts: Kuhn 'The Structure of Scientific Revolutions'; Popper 'Conjectures and Refutations' (selected chapters); Lakatos 'The Methodology of Scientific Research Programmes'; Longino 'Science as Social Knowledge' (selected chapters).

Recommended: Chalmers 'What Is This Thing Called Science?' (optional overview text).

Assignments & Due Dates

Reading Response 1 - Scientific Revolutions - Due: September 22, 2026 (5%)

One-page (400-500 word) response to Kuhn Chapters 1-5. Address: What is a paradigm, and why does Kuhn think normal science is conservative? Where do you agree or disagree with his account? Submit PDF on course portal by 11:59 PM.

Reading Response 2 - Falsificationism and Its Critics - Due: October 6, 2026 (5%)

One-page response to Popper's 'Science: Conjectures and Refutations' and Kuhn's reply. Address: What is Popper's demarcation criterion, and how does Kuhn's account challenge it? Which view do you find more

HPS 220 - History and Philosophy of Science

persuasive, and why?

Essay 1 - Kuhn vs. Popper on Scientific Progress - Due: October 20, 2026 (15%)

3-page argumentative essay (900-1000 words, double-spaced, 12pt Times New Roman, 1-inch margins). Thesis-driven: defend a clear position on whether Kuhn's or Popper's account better explains a specific episode of scientific change. Use at least 3 primary sources and 1 secondary source. Submit PDF on course portal by 11:59 PM.

Reading Response 3 - Lakatos and Research Programmes - Due: November 3, 2026 (5%)

One-page response to Lakatos 'Falsification and the Methodology of Scientific Research Programmes.' Address: How does Lakatos try to reconcile Popper and Kuhn? What is the difference between a progressive and a degenerating research programme?

Essay 2 - Social Epistemology and Scientific Objectivity - Due: November 17, 2026 (20%)

5-page argumentative essay (1400-1600 words). Choose ONE of the following prompts:

- (A) Is Longino right that objectivity requires social diversity in science?
- (B) Does the underdetermination of theory by evidence threaten scientific realism?
- (C) Can feminist philosophy of science improve the epistemic credentials of science?

Engage with at least 4 primary sources and 2 peer-reviewed secondary sources. Submit PDF on course portal.

Reading Response 4 - Science and Society - Due: December 1, 2026 (5%)

One-page response to Oreskes & Conway 'Merchants of Doubt' (excerpt). Address: What does this case study reveal about the relationship between industry, politics, and the production of scientific uncertainty? What are the implications for how we evaluate expert consensus?

Final Paper - Draft - Due: December 4, 2026 (5%)

Submit a complete draft of your final paper (see below) for peer workshop. Bring 2 printed copies to Thursday's class. Draft does not need to be polished but must have a clear thesis, argument structure, and bibliography. Also submit PDF on course portal by 11:59 PM.

Final Paper - A Philosophical Analysis of a Scientific Controversy - Due: December 14, 2026 (30%)

10-page research paper (2800-3200 words). Choose a real scientific controversy (historical or contemporary) and analyze it through the lens of at least TWO philosophical frameworks studied in the course. Your paper must: (1) accurately describe the scientific controversy, (2) apply philosophical frameworks rigorously, (3) defend an original thesis about what the controversy reveals about how science works or should work, and (4) engage with at least 6 academic sources. Submit PDF on course portal by 11:59 PM.

Class Participation - Due: Throughout semester (10%)

Active, prepared, and respectful participation in seminar discussions, assessed holistically at the end of the semester. Quality over quantity. Missing more than 3 classes without excuse results in automatic grade reduction.

HPS 220 - History and Philosophy of Science

Course Schedule Notes

Week 1 (Sep 8-10): What is philosophy of science? Course overview
Week 2 (Sep 15-17): The Scientific Revolution - Galileo, Newton, methodology
Week 3 (Sep 22-24): Kuhn - Structure of Scientific Revolutions I
Week 4 (Sep 29-Oct 1): Kuhn - Revolutions II + Popper on falsification
Week 5 (Oct 6-8): Popper vs. Kuhn; Feyerabend 'Against Method'
Week 6 (Oct 13-15): Lakatos and research programmes
Week 7 (Oct 20-22): Essay 1 due Oct 20; Bayesianism and confirmation theory
Week 8 (Oct 27-29): Scientific realism vs. anti-realism
Week 9 (Nov 3-5): Social epistemology; Longino on objectivity
Week 10 (Nov 10-12): Feminist philosophy of science
Week 11 (Nov 17-19): Philosophy of biology: evolution and function
Week 12 (Nov 24): Science and policy; climate change case study [Tue only]
Week 13 (Dec 1-3): Science denial, expertise, and public trust
Week 14 (Dec 8-10): Final paper workshops; course synthesis
FINAL PAPER DUE: December 14, 2026 (no final exam)

Grading Policy

Reading Responses (4): 20%
Essay 1 (3 pages): 15%
Essay 2 (5 pages): 20%
Final Paper Draft: 5%
Final Paper (10 pages): 30%
Participation: 10%

Late policy: Essays lose 1/3 grade per day late (A -> A- -> B+ ...). Reading responses not accepted late.
Extensions granted only for documented emergencies requested at least 48 hours before the deadline.
Grade scale: A 93+, A- 90, B+ 87, B 83, B- 80, C+ 77, C 73, D 60, F <60.